The climate project

The climate project consists of three parts. In Part 1, you will build a simple carbon cycle model that describes how much of anthropogenic carbon dioxide emissions end up in the atmosphere, the oceans, and the biosphere. In Part 2, you will construct a model to analyze how Earth's energy balance changes due to variations in radiative forcing, for example, from an increased concentration of greenhouse gases in the atmosphere. With this model, you can analyze what happens to the Earth's global mean surface temperature when atmospheric carbon dioxide levels rise. Finally, in Part 3, you will combine Parts 1 and 2 to examine how temperature changes as a result of different carbon dioxide emission scenarios, as well as conduct various analyses using the model.

Project Structure

The project is carried out in groups of two students. For each of the three parts of the project, there will be a supervision session where you can ask questions about the modeling process and discuss how you have solved the tasks and what conclusions you have reached. Completing the tasks and presenting them to the supervisors helps verify that you are on the right track and allows the supervisors to assist if something does not seem correct. **Only the final presentation will be assessed as part of the examination**. This presentation is based solely on Part 3 of the climate project, while Parts 1 and 2 are necessary steps to complete Part 3 and contribute to the understanding of the carbon cycle and Earth's energy balance. See the ”Presentation av Klimatprojektet” page on Canvas for information on presentation structure.
The project can be conducted in any programming language. However, we can only provide guidance on code-specific questions if the code is written in Python and Julia.

Part 1 is more extensive than Parts 2 and 3 of the climate project, so it is important to get started early and complete as much as possible in Part 1 before the first supervision session.

Supervision Plan:

- **Supervision 1:** Questions about the implementation of Part 1 and the project as a whole.
- **Supervision 2:** Review of completed tasks in Part 1 (Tasks 1-7) and questions regarding the implementation of Part 2.
- **Supervision 3:** Review of completed tasks in Part 2 (Tasks 8-10) and questions regarding the implementation of Part 3.

Submission:
Code including clear commenting of code and images for the presentation of your analysis based on your climate model (Tasks 11-13).

Presentation:

- **Presentation of the analysis** based on your climate model (Tasks 11-13).

Grading criteria:

- Pass (0 points): The students obtained the correct model and results and demonstrated an understanding of the code, including the role of each line, and completed the project themselves. Minor careless mistakes are accepted.
- (+1 points): The students have, on the whole, understood the meaning of s, λ, κ, and β, and have reflected on the reference period.
- (+1 points): The students have reflected on how we can think about reducing CO_2 emissions, the role of net negative CO_2 emissions and have also reflected on geoengineering and how these aspects can be connected to sustainable development.

Part 1: Modeling the Carbon Cycle

Background

The most important anthropogenic greenhouse gas is carbon dioxide (CO_2). When analysing the impact of anthropogenic greenhouse gas emissions on the climate, it is crucial to model the rate at which these gases are removed from the atmosphere. Unlike most other gases released into the atmosphere, carbon dioxide does not break down (nor does it fall out with rain or due to gravity like particles do). Instead, it is absorbed by the biosphere and the oceans, where it undergoes bio-chemical reactions and is transformed into other forms of carbon molecules. Over time, the combined atmosphere-biosphere-hydrosphere system reaches an equilibrium with increased carbon content in each subsystem.

This means that not all carbon dioxide emitted disappears from the atmosphere in the foreseeable future—some remains for tens of thousands of years. However, over extremely long time horizons, sediments and rocks play a significant role in the carbon cycle, but we will disregard that here.

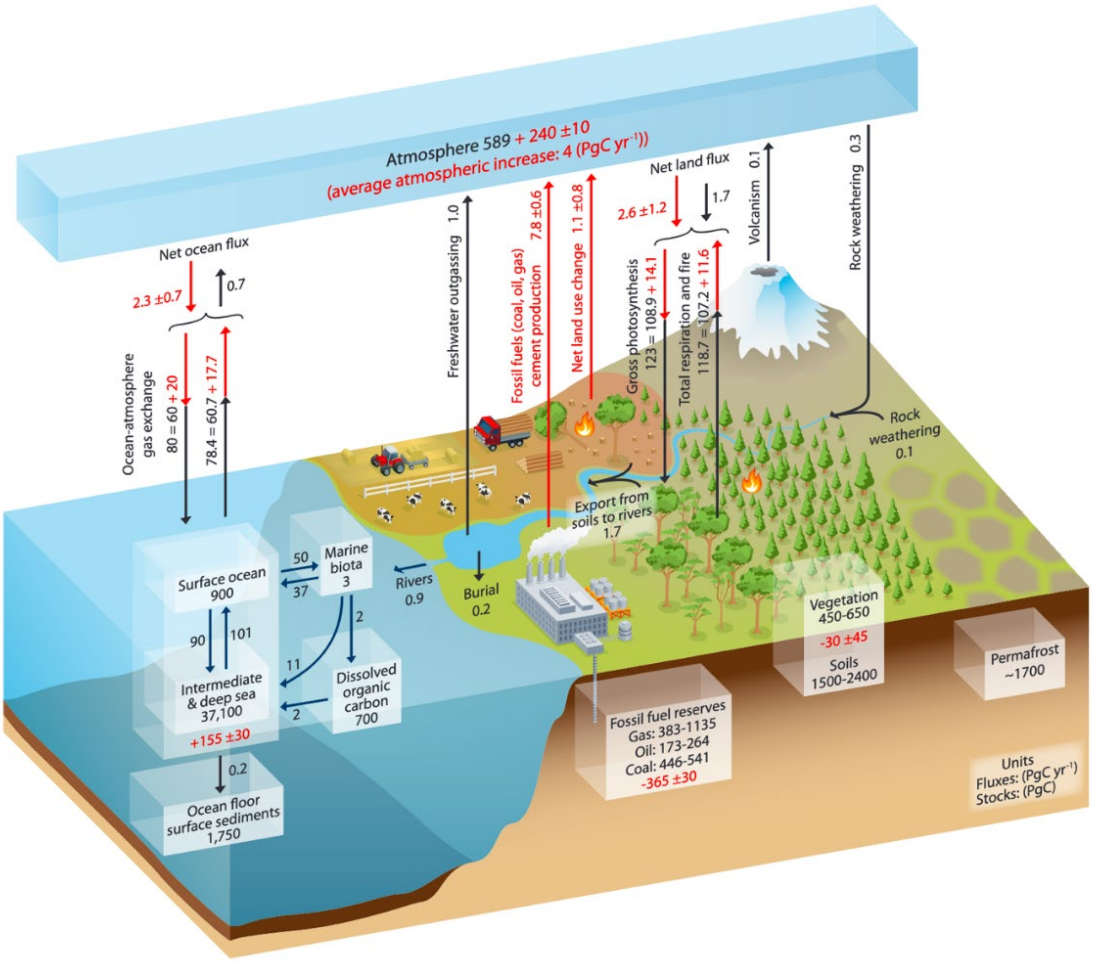

Figure 1: *Carbon Reservoirs and Annual Flows (original from IPCC 2013). The equilibrium state that existed before industrialization is shown in black, while the human impact on the system is highlighted in red.*

Simplified, before deforestation and the use of fossil fuels began, the amount of carbon dioxide in the atmosphere was in balance with the amount of carbon in the biosphere and the oceans. Carbon atoms flowed between the different spheres, and the amount in each sphere remained relatively stable over

time. This balance has now been disrupted due to fossil carbon dioxide emissions and deforestation (see Figure 1).

In this part of the climate project, we will focus on how to model the role of the biosphere and the oceans in the anthropogenically disturbed carbon cycle.

CO_2 Uptake in Vegetation and Oceans

Carbon dioxide does not break down in the atmosphere but is instead absorbed by the oceans (mainly through reactions involving carbonate and bicarbonate in seawater) and by vegetation through photosynthesis. We will model these uptake processes using two different approaches: for the biosphere, we will use a box model, while the ocean's carbon dioxide uptake will be modeled using impulse response functions and convolution.

We will also assess how accurately the combined model of carbon uptake by vegetation and the ocean, when applied to historical carbon dioxide emission estimates, can reproduce observed past atmospheric CO_2 concentrations. Additionally, we will use the model to examine future concentration increases under different emission scenarios.

Biosphere's Uptake of CO_2

The model we will use to describe the carbon content of the biosphere is a box model. The different boxes (B_1-B_3) represent carbon dioxide in the atmosphere, carbon in living biomass above ground, and carbon in the soil (i.e., roots and other soil-bound carbon) (see Figure 2). The pre-industrial sizes of the boxes and the annual flows (F_{ij}) between the boxes (B_i) are given in Figure 2.

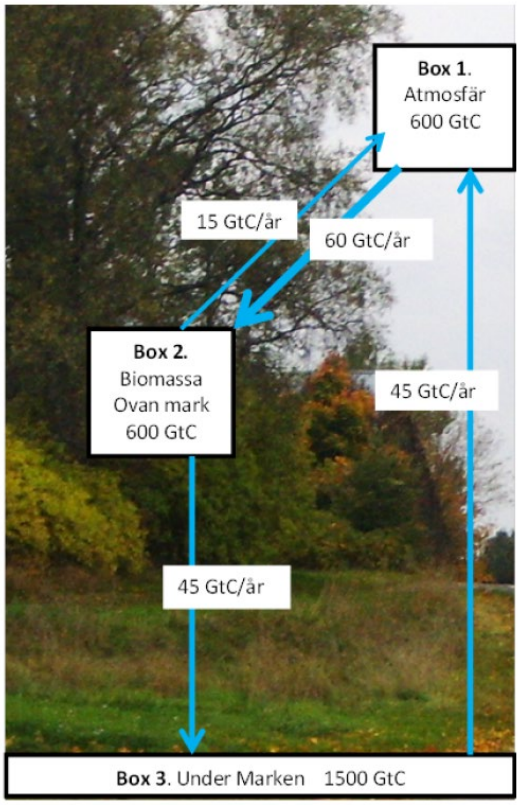

Figure 2: The boxes and the annual flows between them in the terrestrial biosphere and the atmosphere.

Several of these flows are assumed to be linear (which is a reasonable approximation), but photosynthesis, i.e., the flow between the atmosphere and carbon in above-ground biomass, depends

non-linearly on atmospheric carbon dioxide concentration. A good approximation is to assume that photosynthesis has a logarithmic dependence on atmospheric CO_2 levels, meaning that the net primary production (*NPP*) of biomass (resulting from photosynthesis) increases logarithmically with atmospheric CO_2 concentration. The increase in *NPP* with higher CO_2 levels is commonly referred to as CO_2 fertilization or carbon dioxide fertilization (the mechanism is essentially the same as when CO_2 is added to a greenhouse to promote plant growth). In our model, the annual biomass production (i.e., *NPP*), which corresponds to the flow between the atmosphere (box 1) and above-ground biomass (box 2), can be described as follows:

$$NPP = NPP_0 \cdot \left(1 + \beta \cdot ln\left(\frac{B_1}{B_{1,0}}\right)\right). \tag{1}$$

B_1 represents box 1, i.e., the carbon content of the atmosphere in the form of CO_2, and $B_{1,0}$ is the corresponding amount under pre-industrial conditions. Similarly, we represent net primary production with *NPP* and NPP_0. The parameter β is known as the CO_2 fertilization factor, and in a reference case, we assume it to be 0.35. However, there is significant uncertainty in this factor, and it is likely to range between 0.1 and 0.8. In the exercises below, you are encouraged to test different values of this factor.

Using estimates of historical anthropogenic CO_2 emissions (and, for example, a future scenario), the model can calculate annual flows between the different boxes and analyze how the amount of carbon in each box changes over time.

The equations describing the changes in B_i given emissions U, are provided in Equations 2–4. Refer to Figure 2 to understand why the equations are structured as they are.

$$\frac{dB_1}{dt} = \alpha_{31}B_3 + \alpha_{21}B_2 - NPP + U \tag{2}$$

$$\frac{dB_2}{dt} = NPP - \alpha_{23}B_2 - \alpha_{21}B_2 \tag{3}$$

$$\frac{dB_3}{dt} = \alpha_{23}B_2 - \alpha_{31}B_3 \tag{4}$$

The coefficients α_{ij} can be calculated using the values in Figure 2 as follows:

$$\alpha_{ij} = \frac{F_{i,j,0}}{B_{i,0}} \tag{5}$$

Where $F_{ij,0}$ represents the annual flow from box i to box j during pre-industrial times. The coefficients describe the fraction of carbon in box i that flows to box j. α_{ij} is assumed to remain unchanged regardless of variations in the sizes of the boxes (B_i).

The model is most easily solved using a forward Euler method with a time step of 1 year. This approach simplifies the process of integrating this module with the other modules in the climate labs later on.

Task 1: Construct a model for the carbon cycle (in the atmosphere and biosphere) based on the description above. Then, analyze how the flows between the different boxes are affected by the CO_2 emissions scenario in the file *utslappRCP45.csv*, which can be found among the files on CANVAS. To calculate the carbon dioxide concentration, the atmospheric carbon stock must be multiplied by a conversion factor from mass to volume fraction. The conversion factor is 0.469 ppm CO_2/GtC. Compare your calculated CO_2 concentration with the concentration given in

koncentrationerRCP45.csv. Why do you think your calculated concentrations differ from those in *koncentrationerRCP45.csv*?

Task 2: Use the same emissions scenario as in Task 1 and test how varying the CO_2 fertilization factor (β) affects the system. Describe what happens to the atmospheric CO_2 concentration and how the amount of carbon in biomass and soil changes. Explain the results by considering how an increased or decreased fertilization effect influences net primary production (*NPP*), carbon uptake by vegetation, and overall carbon cycling between the atmosphere, biosphere, and soil.

Ocean Uptake of CO_2

The uptake of carbon dioxide by the ocean plays a central role in the carbon cycle. In this part of the project, we will use an impulse response function to represent the ocean's net CO_2 uptake, similar to the approach developed by Maier-Reimer and Hasselmann (1987), which is described in the text *Impulse and Response* (see CANVAS). Note that this text is not part of the examination but is available for those interested and to facilitate understanding of the method used here. The use of impulse response functions in combination with box models provides a simple yet sufficient representation of how CO_2 emissions are removed from the atmosphere.

Although our model is relatively simple, it is on par with models used in applied climate research. In fact, it is at least as advanced and accurate as the carbon cycle model used by William Nordhaus, the recipient of the 2018 Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel, in his research over recent decades.

Using the impulse response function, we can calculate how much of our CO_2 emissions remain in the atmosphere and how much is absorbed by the ocean over time. As previously described, anthropogenic emissions disrupt the natural carbon exchange between different spheres. By implementing an impulse response representation of CO_2 uptake by the ocean, we obtain a model that describes the net flux of carbon from the atmosphere into the ocean. This net flux is computed by convolving the impulse response function with emissions.

From a physical perspective, this flux occurs through a combination of different chemical and physical processes, each with distinct time constants. The exchange of carbon dioxide between the atmosphere and the ocean's surface layer happens relatively quickly, whereas the mixing of carbon molecules into the deep ocean takes hundreds of years.

To represent the different processes that occur on various timescales through which the ocean absorbs carbon dioxide, the impulse response function (transfer function) is expressed as a sum of several exponentially decaying functions, each with different time constants.

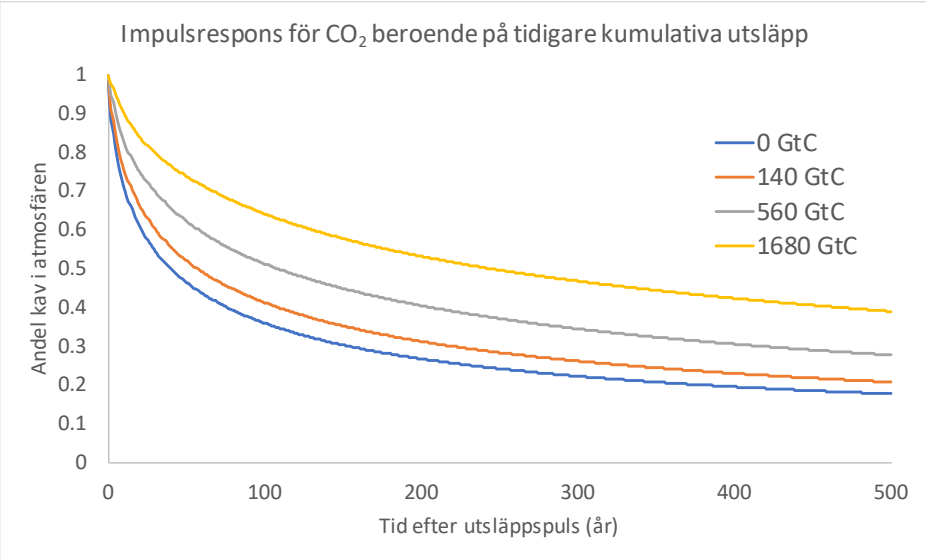

Figure 3: Illustration of how the impulse response depends on cumulative emissions. The fraction of CO_2 remaining in the atmosphere is shown as a function of time after the emission. For reference, the cumulative anthropogenic CO_2 emissions since the start of industrialization have been approximately 700 GtC.

The impulse response *I(t)* is given in its basic form as follows:

$$I(t) = \sum_{i=1}^{n} A_i \cdot e^{-t/\tau_i} \tag{6a}$$

Where A_i represents the fractions of a carbon dioxide emission that decay with a specific time constant τ_i.

In this project, we will use fixed values for the fractions A_i, while the relaxation times τ_i depend on previous cumulative CO_2 emissions (see Equation 7a and Table 1).

The oceans become saturated with carbon dioxide, which reduces their ability to absorb additional CO_2. This effect is illustrated in Figure 3, where the different curves show the impulse response for varying levels of previous cumulative emissions. We capture this dependency by making τ_i a function of past cumulative emissions as follows:

$$\tau_i(t) = \tau_{0,i} \cdot \left(1 + k \cdot \sum_{\bar{t}=0}^{t-1} U(\bar{t})\right) \tag{7a}$$

where $\tau_{0,i}$ is given in Table 1, and the constant $k = 3.06 \cdot 10^{-3}$ per GtC. The value of k is uncertain. A large value of k would mean that the ocean becomes saturated with carbon dioxide more quickly.

Table 1: Fractions A_i and relaxation time constants $\tau_{0,i}$.

i	A_i	$\tau_{0,i}$
1	0.113	2.0
2	0.213	12.2
3	0.258	50.4
4	0.273	243.3
5	0.1430	∞

Task 3: Implement a time-discrete model (based on Equation 6a) that reproduces the impulse responses shown in Figure 3 using the parameter values from Table 1 and Equation 7a. This model should represent how much of an emission pulse remains in the atmosphere over time, depending on previous cumulative emissions.

When we then analyze, for a given year, how much carbon dioxide remains in the atmosphere considering previous annual emissions up to that year, we need to modify Equations 6a and 7a as follows:

$$I(t,\check{t}) = \sum_{i=1}^{n} A_i \cdot e^{-t/\tau_i(\check{t})} \tag{6b}$$

$$\tau_i(\check{t}) = \tau_{0,i} \cdot \left(1 + k \cdot \sum_{\bar{t}=0}^{\check{t}-1} U(\bar{t})\right) \tag{7b}$$

To calculate the amount of carbon dioxide in the atmosphere, $M(t)$, assuming that carbon is only taken up by the ocean (i.e., excluding the biosphere's carbon uptake), we start from a reference year in the past (in this case, the beginning of industrialization) and move forward to the target year for which we want to determine the atmospheric carbon stock. Given the emissions data *U(t)*, the following time-discrete convolution is used:

$$M(t) = M_0 + \sum_{\tilde{t}=0}^{t} I(t-\tilde{t},t) \cdot U(\tilde{t}) = M_0 + \sum_{\tilde{t}=0}^{t} \left(\sum_{i=1}^{n} A_i \cdot e^{-(t-\tilde{t})/\left(\tau_{0,i}\cdot\left(1+k\cdot\sum_{\bar{t}=0}^{\tilde{t}-1} U(\bar{t})\right)\right)} \right) \cdot U(\tilde{t}) \tag{8}$$

Where M_0 represents the carbon stock in the atmosphere before industrialization (see Figure 2).

Task 4: The next step is to implement a model based on Equation 8 and run it using the emissions data from the file *utslappRCP45.csv* to calculate how atmospheric CO_2 concentration would have developed if carbon were only taken up by the ocean. To compute the CO_2 concentration, the atmospheric carbon stock (i.e., the total mass C in CO_2) must be multiplied by a conversion factor from mass to volume fraction in the atmosphere, with a value of *0.469 ppm CO_2/GtC.* Compare your calculated concentrations with those given in *koncentrationerRCP45.csv*.

Biosphere + Ocean Uptake of CO_2

Now, we will integrate the box model for the biosphere's carbon uptake with the impulse response model for the ocean's carbon uptake.

Task 5: Draw a new box model similar to the one in Figure 2, but in addition to the carbon flows between the biosphere and atmosphere, include the net uptake of carbon by the ocean according to the impulse response model. Also, add anthropogenic emissions to the diagram. There is no need to quantify the flows now — just illustrate the structure conceptually.

Task 6: Connect the impulse response model for oceanic CO_2 uptake with the box model for biospheric CO_2 uptake. Here, you need to replace the emissions term *U* in Equation 8 with the right-hand side of Equation 2, meaning anthropogenic emissions minus the net CO_2 uptake in the terrestrial carbon cycle. Keep in mind that *M* in Equation 8 and B_1 in Equation 2 both represent the atmospheric carbon stock. Then, calculate the atmospheric CO_2 concentration using the emissions data from *utslappRCP45.csv* and compare the model results with the harmonized values provided in *koncentrationerRCP45.csv*. Adjust β so that the model-calculated concentration aligns reasonably well with the observed values.

Task 7: Analyze the long-term fate of anthropogenic CO_2 emissions. How do the different carbon stocks (atmosphere, biomass, soil, and ocean) change over time until the year 2100? How do the values in 2100 depend on the assumptions for β and k? How do the results change if the assumptions

for β and k are halved or doubled? Try to explain the changes in the results based on the underlying mechanisms.

Part 2: Energy balance modelling

Background

In this section, we will calculate the temperature response to human activities that influence Earth's radiative balance (see Goosse et al., sections 2.1.2 and 4.1.1–4.1.4 for background information). The temperature response in this context refers to the increase in the global mean temperature at the surface of land and oceans relative to pre-industrial levels.

Earth's average temperature is a key indicator of changes in the climate system as a whole. In international climate negotiations within the UN framework, this is the primary metric referenced. For example, in the Paris Agreement, it was established that efforts should be made to "… keep the increase in global mean temperature well below 2°C above pre-industrial levels and pursue efforts to limit the temperature increase to 1.5°C above pre-industrial levels, recognizing that this would significantly reduce the risks and impacts of climate change…"

To estimate the temperature response to anthropogenic greenhouse gas emissions and other climate-forcing agents, we will first calculate the contribution to radiative forcing (see Figure 4 below and Goosse et al., sections 4.1.1–4.1.2). Radiative forcing measures the warming effect resulting from changes in greenhouse gas concentrations and other climate-affecting substances. Using radiative forcing and an energy balance model, we can determine its effect on the global average surface temperature.

In the energy balance model, the change in Earth's energy balance (relative to pre-industrial levels) is calculated based on radiative forcing, outgoing thermal radiation from Earth, and the energy required to heat the ocean above its pre-industrial temperature.

If the ocean had no heat capacity, Earth's outgoing thermal radiation and radiative forcing would essentially be in sync, meaning that global temperature would instantly follow changes in radiative forcing. However, due to the ocean's large heat capacity, temperature changes on Earth occur more gradually in response to changes in radiative forcing. Heating large volumes of water takes time.

In the model, the combined effective heat capacity of the upper ocean, atmosphere, and land surface is represented as a single homogeneous upper box, while the deep ocean is modeled as a separate homogeneous lower box. The upper and lower boxes exchange energy, with the rate of exchange being proportional to the temperature difference between them. Although simple, this model effectively captures the key timescales of the global climate system.

In the model, climate-forcing agents affect the upper box through their radiative forcing, and heat is then transferred between the upper and lower boxes (see Figure 6 below for an illustration).

This type of model has been widely used in scientific research over the past 40 years. It is particularly useful for understanding the temperature impact of different emissions scenarios or for analyzing key uncertainties in the climate system, such as climate sensitivity.

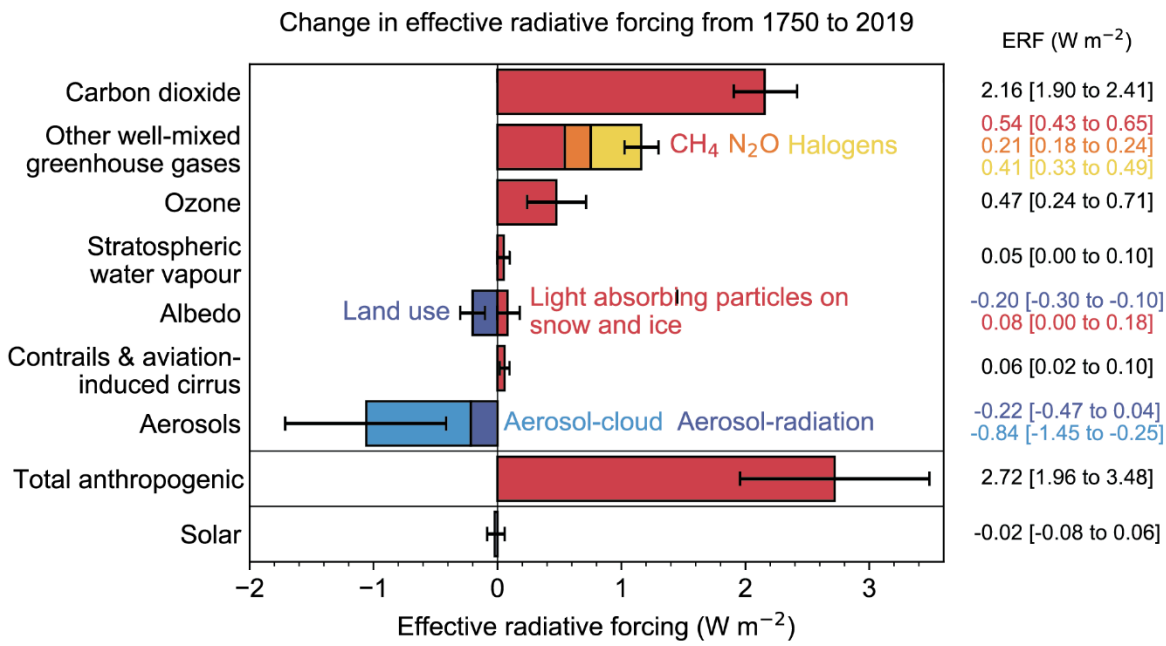

Figure 4: Radiative forcing with uncertainties distributed across climate-affecting substances and changes in solar radiation (according to IPCC 2021).

Radiative forcing

Radiative forcing is defined as the change in net radiation at the tropopause (which can practically be interpreted as the top of the atmosphere) following a radiative disturbance, such as an increase in greenhouse gas concentrations, assuming a fixed temperature in the troposphere. This measure has been developed to estimate the warming effect of climate-affecting substances and to facilitate the study of climate change. In this project we use the concepts of effective radiative forcing and radiative forcing interchangeably, although there are scientific differences in the concepts.

Simple empirical mathematical relationships are often used to describe the connection between the concentration of a greenhouse gas and its impact on radiative forcing. These relationships are curve fits to results from detailed radiative balance models, which are based on Planck's radiation law. These models calculate the absorption and emission of radiation in the atmosphere at different wavelengths, using high resolution and under different assumptions about the concentration of the gas in question.

The approximated relationships between radiative forcing and concentration differ for various greenhouse gases. This is because different substances exhibit varying degrees of saturation in the atmosphere and absorb thermal radiation at different wavelengths.

For carbon dioxide's effect on radiative forcing, we will use a simple relationship that describes the connection between gas concentration and its impact on radiative forcing. This is expressed in the following equation:

$$RF_{CO_2} = 5.35 \cdot ln\left(\frac{p_{CO_2}}{p_{CO_{2,0}}}\right) \tag{9}$$

As seen, the contribution of carbon dioxide to radiative forcing increases logarithmically with the relative deviation from the pre-industrial level. p_{CO_2} represents the concentration at the time for which

the radiative forcing is being calculated, while $p_{CO_{2,0}}$ is the concentration before industrialization began. The reason that carbon dioxide's impact can be described with a logarithmic relationship is due to saturation in the wavelength bands where carbon dioxide absorbs thermal radiation (see Figure 5).

Radiative forcing from other greenhouse gases and aerosols in Figure 4 should be included in the model by using two predefined time series from RCP45 (a standardized emissions scenario), one for aerosols and one for the remaining climate-affecting substances' radiative forcing. These are available in the file *radiativeForcingRCP45.csv*. The sum of the different radiative forcing contributions gives the total radiative forcing.

Task 8: Construct a radiative forcing module based on the description above. The value of p_{CO_2} (i.e., the partial pressure or concentration) can be obtained from your carbon cycle model or from the concentration values in *koncentrationerRCP45.csv*. Calculate and visualize the radiative forcing for CO_2, and compare these values with the CO_2 radiative forcing in *radiativeForcingRCP45.csv*.

Task 9: Sum the radiative forcing for other climate-affecting substances and aerosols, which can be found in *radiativeForcingRCP45.csv*. Multiply the radiative forcing for aerosols by a scaling factor, *s*, with a default value of 1. In this part of the project, *s,* will be set to 1, but in the final part (part 3), this value will be varied. The reason for using this scaling factor is the large uncertainty surrounding the cooling effect of aerosols.

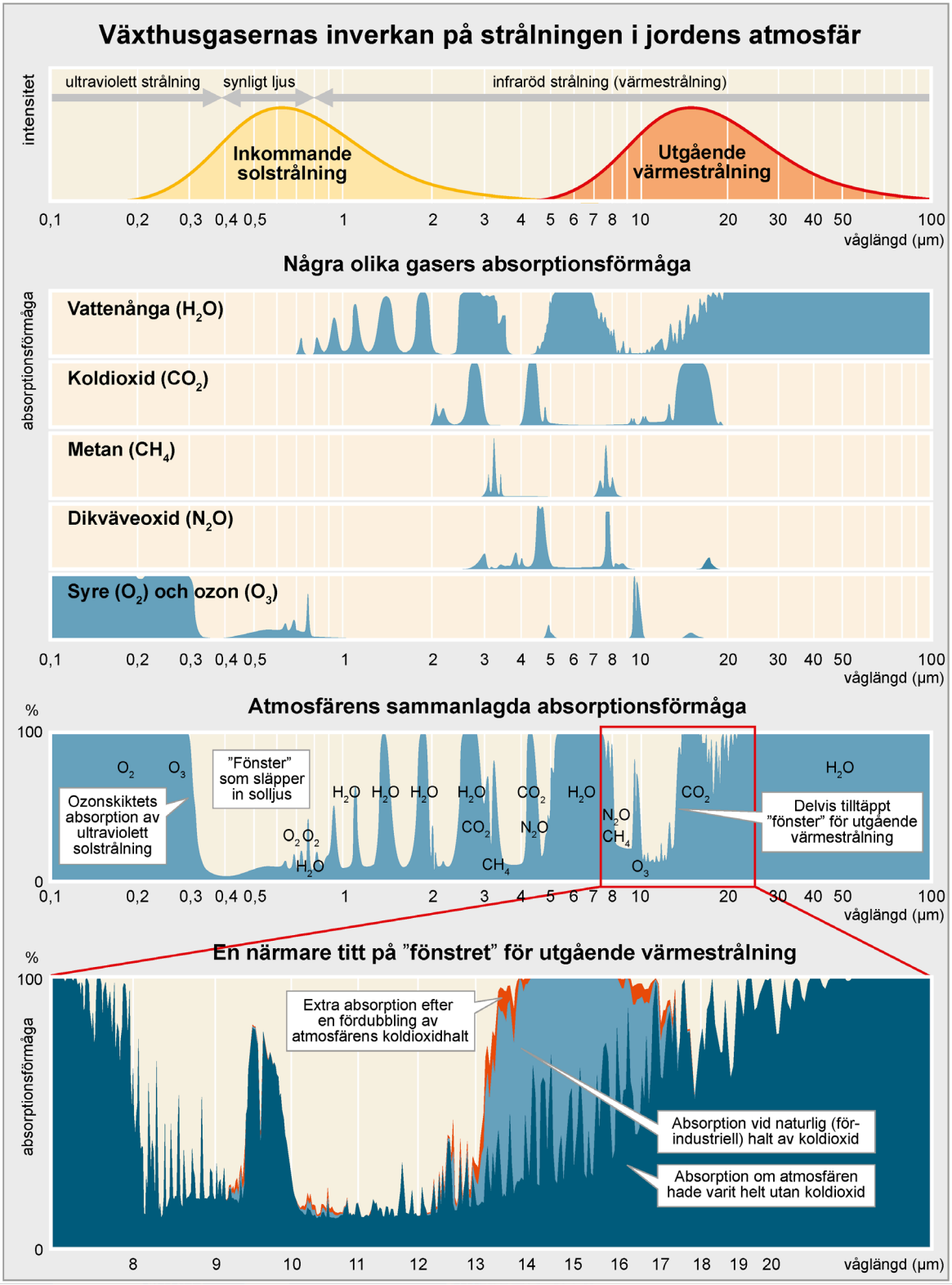

Figure 5: Absorption and emission across different wavelengths in the atmosphere (according to Piexoto and Oort 1992, with updated data from MODTRAN, graphical adaptation by Bernes 2016).

A two-box energy balance model

The temperature response is calculated using a simple energy balance model; see Goosse et al., chapter 4.1.4, for a similar but more simplified model.

The model parameters are adjusted so that the model's temperature response exhibits a surface temperature response (ΔT_1) similar to that produced by more sophisticated climate system models. These more advanced models explicitly simulate the processes driving heat exchange between different ocean depths as well as between the ocean, atmosphere, and land. Here, we simplify this by describing the average temperature of the upper ocean (including land and atmosphere) as a homogeneous and well-mixed box, while the temperature evolution of the deep ocean (ΔT_2) is represented as another linked, well-mixed box.

Equation 10 describes the heat flux between the upper box and the deep ocean box. Both equations are derived from the principles of energy conservation and Fourier's law of heat conduction.

The changes in surface temperature ΔT_1 and deep ocean temperature ΔT_2 are therefore given by the following equations:

$$C_1 \frac{d\Delta T_1(t)}{dt} = RF(t) - \frac{\Delta T_1(t)}{\lambda} - \kappa \cdot \left(\Delta T_1(t) - \Delta T_2(t)\right) \tag{10}$$

$$C_2 \frac{d\Delta T_2(t)}{dt} = \kappa \cdot \left(\Delta T_1(t) - \Delta T_2(t)\right) \tag{11}$$

Where RF is the net radiative forcing, ΔT_1 is the temperature increase [K] since the pre-industrial level in the upper box, ΔT_2 is the temperature increase [K] since the pre-industrial level in the deep ocean box, $C_1$is the effective heat capacity of the upper box [W·yr·K^{-1}·m^{-2}],C_2 is the effective heat capacity of the deep ocean box [W·yr·K^{-1}·m^{-2}], κ is the exchange coefficient between the upper box and the deep ocean box [W·K^{-1}·m^{-2}], and finally, λ is the climate sensitivity parameter [K·W^{-1}·m^2].

At the pre-industrial level, RF, ΔT_1, and ΔT_2 were zero, meaning that at $t = 0$, the conditions were $\Delta T_1(0) = 0$ and $\Delta T_2(0) = 0$.

The reason this model is called an energy balance model is that it describes the energy flow between the boxes per unit time. If we consider Equation 10 above, it can be interpreted in terms of energy exchange as follows:

$C_1 \frac{\partial d(t)}{dt}$ – Energy uptake per unit time in the upper box.

$RF(t)$ – Radiative forcing.

$\frac{\Delta T_1(t)}{\lambda}$ – Outgoing thermal radiation to space.

$\kappa \cdot (\Delta T_1(t) - \Delta T_2(t))$ – Energy uptake per unit time in the deep ocean box.

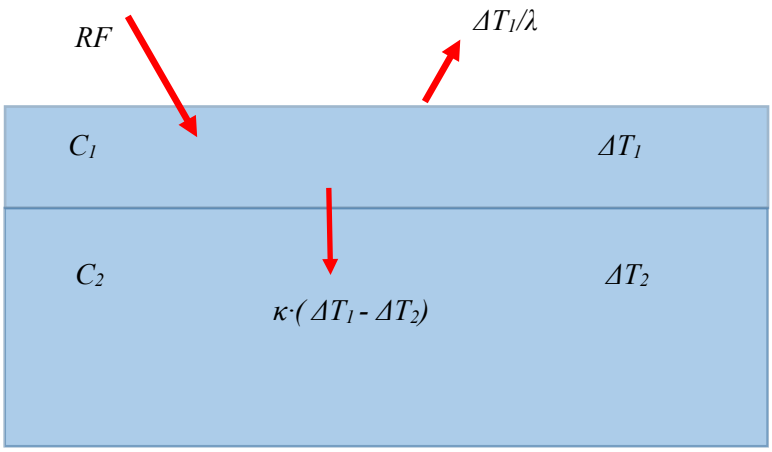

Figure 6. Illustration of the two-box energy balance model.

Task 10 Discretize equations 10 and 11 using a forward Euler method with an annual time step. Use the discretized versions of equations 10 and 11 along with the parameter values below to simulate the energy balance model.

a) Test the model by analyzing the temperature response based on a radiative forcing step of 1 W/m^2, meaning that radiative forcing increases from 0 to 1 W/m^2 in the first time step. What is the simulated equilibrium temperature? Verify this numerically by setting the model's time horizon sufficiently long (the equilibrium temperature is given by är $\Delta T_1 = \Delta T_2 = RF \cdot \lambda$, which should be evident from equations 10 and 11).
b) Analyze the effect of the climate sensitivity parameter λ and the exchange coefficient κ on the time required to reach equilibrium temperature. The transient phase, i.e., the period before the system reaches equilibrium, can be analyzed by calculating the "e-folding time," which is the time required for the temperature response to reach $(1-e^{-1})$ of its equilibrium value for different values of λ and κ. Reasonable values for λ range between 0.5 and 1.3 $[K \cdot W^{-1} \cdot m^2]$, and for κ between $[W \cdot K^{-1} \cdot m^{-2}]$. Try to intuitively understand the results and why ΔT_1 and ΔT_2 evolve as they do, depending on the choice of λ and κ.
c) Analyze the energy fluxes (radiative forcing, ocean heat uptake, and outgoing thermal radiation to space) for the step and the parameter values used in task 10b. How do ocean heat uptake and outgoing radiation to space depend on λ and κ? Does the law of energy conservation hold (i.e., is incoming energy minus outgoing energy equal to the accumulated energy in the system)? Analyze the fluxes and balance over a 200-year time horizon and try to intuitively interpret the results.

Tip: When implementing the numerical model, be thorough with unit analysis.

Parameter values:

λ = Climate sensitivity parameter= 0.8 $[K \cdot W^{-1} \cdot m^2]$, uncertainty span 0.5-1.3 $[K \cdot W^{-1} \cdot m^2]$.
κ = Heat exchange coefficient=0.5 $[W \cdot K^{-1} \cdot m^{-2}]$, uncertainty span 0.2- 1 $[W \cdot K^{-1} \cdot m^{-2}]$.
RF = Radiative forcing $[W/m^2]$.
c = The specific heat capacity of water = 4186 (J/kg)/K.
ρ = The density of water =1020 kg/m^3
h = The effective depth of the surface box = 50 m.
$C_1 = c \cdot h \cdot \rho$ = The effective heat capacity of the surface box. You should recalculate so that the parameter has the unit $[W \cdot yr \cdot K^{-1} \cdot m^{-2}]$.
d= The effective depth of the deep ocean box = 2000 m.
$C_2 = c \cdot d \cdot \rho$ = The effective heat capacity of the deep ocean box. You should recalculate so that the parameter has the unit $[W \cdot yr \cdot K^{-1} \cdot m^{-2}]$.

Part 3: Finalization and Analysis of the Climate Model

In this final part of the climate project, you will integrate the different modules and use the model to analyze a series of questions.

When assembling a model composed of multiple smaller models (modules), it is important to ensure consistency in naming conventions and variable usage. It is also crucial to have completed the previous tasks and verified that each sub-model functions correctly.

Task 11 In this task, you will combine the carbon cycle model with the energy balance model. When doing this, consider that the time discretization may differ between the two modules depending on your previous modeling choices.

Use parameter values for the carbon cycle from Part 1 and for the energy balance model from Part 2. Calculate the increase in global mean surface temperature for the period 1765–2024 based on historical global CO_2 emissions and the historical RF estimates for the other substances *radiativeForcingRCP45.csv*. Compare your results with NASA's estimated values for global temperature anomalies (see *NASA_GISS.csv*). NASA's time series shows the temperature increase over the period 1880–2019 relative to a reference period based on the average temperature from 1951 to 1980. Adjust your results so they are easily comparable to NASA's dataset.

a) How does the choice of reference period affect the results? What would be an appropriate reference period if the goal is to describe temperature changes relative to pre-industrial times?

b) Test different values for the climate sensitivity parameter λ (0.5, 0.8, and 1.3 K/W/m²). What happens? Adjust the exchange coefficient κ (for heat transfer between the surface and deep ocean) and the scaling factor $\boldsymbol{s}$ (for aerosol forcing) to find a model response that closely matches the observed temperature anomaly from NASA for each of the three λ values. What values of κ and s provide a good fit to NASA's temperature series for each assumption of climate sensitivity?

c) The parameters λ, κ, and s all have uncertainties. Discuss the feasibility of statistically estimating these values from global temperature time series to reduce uncertainty intervals.

Task 12: Test different future CO_2 emission scenarios along with a scenario for radiative forcing from other climate-affecting substances, based on *radiativeForcingRCP45.csv*, a "middle of the road" scenario that we will use in combination with various CO_2 emission scenarios.

a) Use a climate sensitivity parameter of $\lambda = 0.8$ K/W/m² along with the values of κ and s that provided a good fit to the historical temperature anomaly in Task 11. Generate and present temperature projections for the period 1765–2100 under the following three emissions scenarios (see Figure 7):
i. CO_2 emissions decrease linearly to zero by 2070, continue decreasing at the same annual rate until 2100. After 2100, emissions remain constant at the level obtained in 2100 until the end of the model's time horizon (2200).
ii. CO_2 emissions remain constant at today's level until the end of the model's time horizon.
iii. CO_2 emissions increase linearly until 2100, reaching 200% of today's emissions. After 2100, emissions remain constant at that level.

b) How much does the temperature increase over the century relative to pre-industrial levels in the three cases i–iii? Illustrate in a figure.

c) How large is the spread in temperature responses by 2100 in the three cases i–iii, given the uncertainty in the climate sensitivity parameter λ (along with estimates of *s* and κ that provide a good fit to historical temperatures)? Illustrate in a figure.

d) As you may now observe, temperature projections for 2100 can vary significantly due to both uncertainties in the climate system (mainly due to *λ*, but also *s* and *κ*) and our choices regarding CO_2 emissions. Given this uncertainty, how should we think about how much CO_2 emissions should be reduced in the coming decades to meet the climate goals of the Paris Agreement? Use the Konjunkturrådets report and texts on sustainable development as inspiration for your reasoning.

e) The low emissions scenario has negative CO_2 emissions towards the end of the century. How can such negative CO_2 emissions be materialised? Negative CO_2 emissions tend to lead to a peak and decline in global mean surface temperature (make a plot to verify that this is the case). In what way are such temperature pathways relevant for reaching the temperature goals stated in the Paris Agreement?

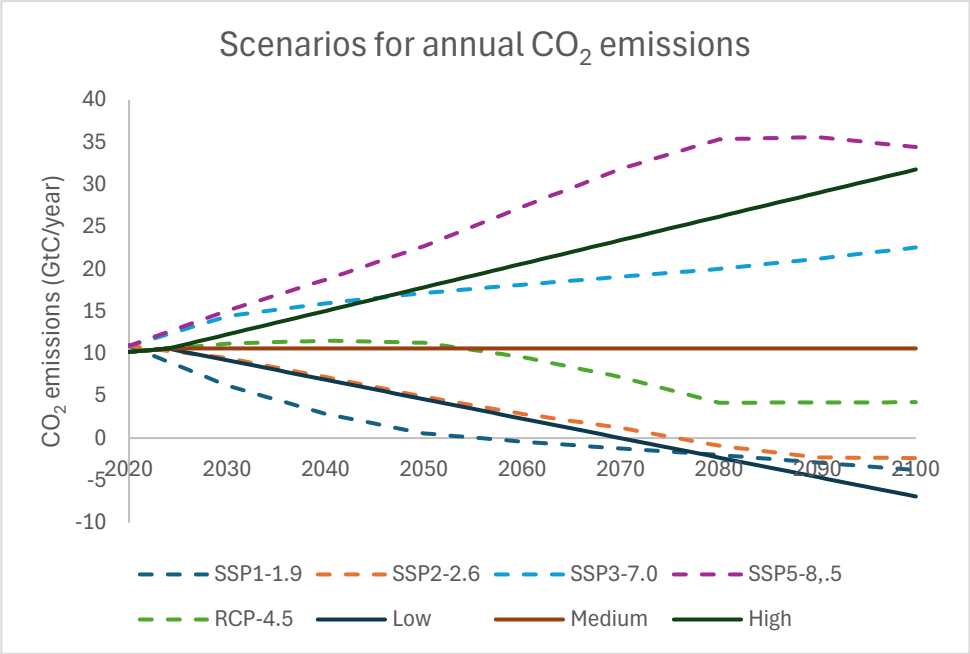

Figure 7. Scenarios for the development of anthropogenic CO_2 emissions during the current century. The SSP and RCP scenarios are standard in climate modeling, while our low, medium, and high scenarios correspond to our scenarios i, ii, and iii.

Task 13: Let us assume that future emissions of climate-affecting substances will be high (following scenario iii from the previous task), meaning a world where nations do not collaborate significantly to reduce greenhouse gas emissions and where fossil fuel availability remains abundant. Under these circumstances, geoengineering is discussed as a possible alternative to mitigate global climate change.

One such geoengineering approach could involve dedicated emissions of aerosol-forming gases in the stratosphere, where the aerosols would reflect incoming solar radiation and thus cool the Earth. This approach could have significant consequences for the climate and would require continuous aerosol emissions, as their lifespan in the stratosphere is only a few years. If aerosol emissions were to stop, their cooling effect would disappear within a few years.

a) <u>Simulating the Termination of Geoengineering</u>. Analyze what would happen if geoengineering were implemented for a period but then, for some reason, abruptly stopped. Test a scenario where incoming solar radiation is reduced by 4 W/m² from 2050 to 2100. What would happen to global mean temperature in such a scenario (over the entire period up to 2200)?

b) Geoengineering from a Sustainable Development Perspective. Critically discuss the feasibility of geoengineering as a climate change response from the perspective of sustainable development. In your discussion, compare geoengineering with strategies for reducing CO_2 emissions in the energy system, including the role of negative CO_2 emissions. Evaluate the extent to which geoengineering is consistent with long-term climate objectives and the core principles of sustainable development. Draw on course materials, including the texts on sustainable development, to frame and support your analysis.